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Anti-obesity effects of *Celosia cristata* flower extract *in vitro* and *in vivo*

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ABSTRACT

Background: The overstoring of surplus calories in mature adipocytes causes obesity and abnormal metabolic activity. The anti-obesity effect of a *Celosia cristata* (CC) total flower extract was assessed *in vitro*, using 3T3-L1 pre-adipocytes and mouse adipose-derived stem cells (ADSCs), and *in vivo*, using high-fat diet (HFD)-treated C57BL/6 male mice.
Methods: CC extract was co-incubated during adipogenesis in both 3T3-L1 cells and ADSCs. After differentiation, lipid droplets were assessed by oil red O staining, adipogenesis and lipolytic factors were evaluated, and intracellular triglyceride and glycerol concentrations were analyzed. For *in vivo* experiments, histomorphological analysis, mRNA expression levels of adipogenic and lipolytic factors in adipose tissue, blood plasma analysis, metabolic profiles were investigated.
Results: CC treatment significantly prevented adipocyte differentiation and lipid droplet accumulation, reducing adipogenesis-related factors and increasing lipolysis-related factors. Consequently, the intracellular triacylglycerol content was diminished, whereas the glycerol concentration in the cell supernatant increased. Mice fed an HFD supplemented with the CC extract exhibited decreased HFD-induced weight gain with metabolic abnormalities such as intrahepatic lipid accumulation and adipocyte hypertrophy. Improved glucose utilization and insulin sensitivity were observed, accompanied by the amelioration of metabolic disturbances, including alterations in liver enzymes and lipid profiles, in CC-treated mice. Moreover, the CC extract helped restore the disrupted energy metabolism induced by the HFD, based on a metabolic animal monitoring system.
Conclusion: This study suggests that CC total flower extract is a potential natural herbal supplement for the prevention and management of obesity.

1. Introduction

Obesity represents a significant global health challenge characterized by abnormal and excessive fat distribution throughout the body, contributing to increased weight gain [1]. This condition is significantly influenced by various factors, encompassing genetics, endocrinological aspects, environmental influences, excessive caloric intake, and lifestyle choices [2]. The repercussions of obesity extend to various health complications, including osteoarthritis, cardiovascular diseases, type 2 diabetes, hypertension, and certain types of cancer [3,4]. According to

Abbreviations: AA, antibiotic–antimycotic; ADSCs, adipose-derived stem cells; ALT, alanine aminotransferase; AST, aspartate aminotransferase; CC, *Celosia cristata*; DMEM, Dulbecco's modified Eagle's medium; EE, energy expenditure; FBS, fetal bovine serum; GC, *Garcinia cambogia*; H&E, hematoxylin and eosin; HDL, high-density lipoprotein; HFD, high-fat-diet; HPLC, high-performance liquid chromatography; ITT, insulin tolerance test; KMP, kaempferol; LDL, low-density lipoprotein; ND, normal diet; OGTT, oral glucose tolerance test; ORO, Oil red O; RER, respiratory exchange ratio; TG, triacylglycerol; VO_2, oxygen consumption; VCO_2, CO_2 expiration.

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the World Health Organization report in 2022, the prevalence of obesity is staggering, with more than 890 million adults worldwide categorized as obese and a larger population of 2.5 billion considered overweight adults (18 years and older) [5], implying a troublingly increasing trend over time.

In obesity, dysregulation of the balance between adipogenesis and lipolysis disrupts the metabolic equilibrium [6]. Adipocytes, engaging in adipogenesis, accumulate energy as lipids/triacylglycerols (TGs) [7], whereas during lipolysis, TGs are degraded into free fatty acids and glycerol [8]. Measurements of intracellular TG content and extracellular glycerol released from cells and tissues provide important insights into obesity states [9]. During adipogenic differentiation from mesenchymal stem cells into lipid-storing adipocytes, specific transcription factors and enzymes, such as peroxisome proliferator-activated receptor gamma (Pparg), CCAAT/enhancer-binding protein α (Cebpa), fatty acid binding protein 4 (Fabp4), and fatty acid synthase (Fasn), enhance adipocyte differentiation, leading to obesity-related risks [10]. Conversely, lipolysis, involving enzymes such as hormone-sensitive lipase (Hsl) [11], acyl-coenzyme A oxidase 1 (Acox1) [12], and carnitine palmitoyltransferase 2 (Cpt2) [13], facilitates TG breakdown for energy production through β-oxidation. Disrupted anabolism and catabolism in adipocytes can contribute to metabolic complexities such as dyslipidemia, hepatic steatosis, and elevated plasma insulin levels [14]. Targeting the modulation of adipogenesis and/or lipolysis has thus emerged as a fundamental strategy for therapeutic approaches aimed at managing obesity.

Phytochemicals can exert regulatory effects across adipocyte development stages by promoting the transformation of white fat cells into a browning phenotype [15], increase mitochondrial biogenesis and fat oxidation [16]. This process is controlled by various factors, including browning markers such as peroxisome proliferator-activated receptor-gamma co-activator-1a (Pgc1a) [17], uncoupling protein1 (Ucp1), and PR domain containing 16 (Prdm16). Their expression in white adipose tissue (WAT) stimulates browning phenotypes [18]. Mitochondrial biogenesis markers, including cytochrome c oxidase 4b (Cox4b), a major component of the mitochondrial respiratory chain, serves as the terminal enzyme known as cytochrome c oxidase (Cox). It facilitates the transfer of electrons from reduced cytochrome c (Cytc) to oxygen, leading to the conversion of oxygen into water [19]. Additionally, β-oxidation marker: medium chain acyl-Coenzyme a dehydrogenase (Mccad) plays crucial role in mitochondrial fatty acid β-oxidation by catalyzing its initial step [20]. These factors collectively contribute to the regulation of adipocyte metabolism and mitochondrial energy homeostasis.

Although dietary plans and lifestyle alterations are commonly recommended for obesity management, their adherence is challenging, leading to limited efficacy and a low success rate in practical applications [21]. Consequently, there is a demand for therapeutic interventions that address obesity with concurrent dietary modifications. Conventional pharmaceuticals, including lorcaserin (which reduces appetite) and orlistat (which reduces fat absorption), have been developed; however, there are concerns regarding long-term safety and the associated significant side effects, including diarrhea and cardiovascular complications [22,23]. As a result, there is marked interest in natural medicinal plants and herbal supplements owing to their bioactive compounds and relatively fewer adverse effects. Natural alternative sources are rich in diverse bioactive compounds, have fewer complexities, emphasize sustained safety, and have a history of traditional use for various health issues [24]. Therefore, they have attracted attention for their persistent safety and potential therapeutic benefits, which have increased global research into herbal remedies and supplements for conditions including obesity.

Celosia cristata (CC), recognized as a medicinal plant, has important applications, serving as a nutritional source, medicinal remedy, and herbal supplement [25]. Traditionally, dried CC flowers have been utilized to alleviate disorders such as osteoporosis, hematuria, and abnormal menstrual bleeding, owing to their purported properties [26,27]. Beyond their historical use, studies have revealed their beneficial effects, including antioxidant, hepatoprotective [28], anti-microbial [29], and anti-aging [30] activities. Additionally, CC was found to influence the adipogenic potential of human adipose tissue progenitor cells during their commitment and differentiation toward the adipogenic lineage *in vitro* [31]. Recent research efforts have focused on exploring various phytochemicals, such as flavonoids, tannins, and phenolic compounds derived from medicinal plants, flowers, and herbs, for their anti-obesity properties [32]. The various groups of phytochemicals have been identified in the ethanolic extract of CC, such as phenolic acids (4-hydroxyphenethyl alcohol), flavonols (kaempferol, quercetin), phytosterols (β-sitosterol, 2-hydroxy octadecanoic acid, stigmasterol), and isoflavones (cochliophilin A, cristatein) [33,34]. Among these phytochemicals, flavonoids are the main compounds of the ethanolic extract of CC [34]. Kaempferol (KMP), a well-known flavonoid compound found in numerous dietary sources, exerts anti-obesity effects by inhibiting adipogenesis and increasing lipolysis in 3T3-L1 cells and promoting lipolysis and alleviating obesity-related pathology in high fat diet induced obese mice [35,36]. While KMP, a major constituent extracted from CC, has been identified [25], the specific effects of the ethanolic total flower extract of CC, related to the inhibition of body weight gain conditions of obesity induced by dietary factors, and the underlying mechanisms, remain unclear.

In present study, we aimed to investigate the anti-obesity effect of CC using mouse 3T3-L1 pre-adipocytes and primary adipose-derived stem cells (ADSCs) *in vitro* and an HFD-treated mouse model *in vivo*.

2. Material and methods

2.1. Plant materials and extract preparation and characterization

CC flower extract powder was provided by NINE B Co., Ltd. (Daejeon, Republic of Korea). The CC flowers were weighed and subjected to extract preparation. Dried flowers were soaked in 30% ethanol and re-extracted using the same solvent. All extracts were collected and then filtered using a polypropylene filter (The Zone Filter, Gyeonggi, Republic of Korea) to remove any non-soluble substances. The filtrate was concentrated in a vacuum evaporator and spray-dried with dextrin. The extracted CC was used for sample preparation, along with high-performance liquid chromatography (HPLC; Agilent Technologies, Santa Clara, CA, USA), and the identification of KMP.

2.2. Cell culture and adipogenic induction

Mouse fibroblast 3T3-L1 pre-adipocyte cells (less than 13 passages) were purchased from the Korean Cell Line Bank (KCLB No. 10092.1), cultured, and maintained in high-glucose Dulbecco's modified Eagle's medium (DMEM; Invitrogen, Carlsbad, CA, USA) with 10% bovine calf serum (Invitrogen, Carlsbad, CA, USA) and 1% antibiotic–antimycotic (AA; Invitrogen). For the induction of adipogenesis, cells were incubated with DMEM containing 10% fetal bovine serum (FBS; Invitrogen), 1% AA, 1 μM dexamethasone, 0.5 mM 3-isobutyl-1-methylxanthine, and 10 μg/mL insulin (together referred to as MDI) for 3 days, followed by a 5-day incubation with DMEM containing 10% FBS, 1% AA, and 10 μg/mL insulin. The mouse ADSCs were obtained from the stromal vascular fraction of 6-week-old mice (C57BL/6). Briefly, epidydimal adipose tissue isolated from the mouse was digested with type II collagenase for 1.5 h, and the reaction was stopped using low-glucose DMEM with 10% FBS. Digested adipose tissue was filtered with a 100-μm cell strainer (Corning, NY, USA) and centrifuged at 2,500 rpm for 10 min. Primary cultured ADSCs were directly cultured in low-glucose DMEM containing 10% FBS and 1% AA, and adipogenesis was induced in ADSCs using adipogenic differentiation medium (Mesencult™, STEMCELL Technologies, Vancouver, BC, Canada) for 8 days.

2.3. Cell viability test

3T3-L1 cells or ADSCs were treated with different concentrations (10, 20, 50, and 100 μg/mL) of CC in a culture medium and incubated for 8 days, and viability was assessed using the D-Plus™ CCK cell viability kit (Dongin Biotech, Seoul, Republic of Korea) at 450 nm using an iMark™ Microplate Absorbance Reader (Bio-Rad, Hercules, CA, USA).

2.4. Oil-red O (ORO) staining

After adipocyte differentiation, cells were washed with PBS and fixed using 4% paraformaldehyde (BIOSESANG, Seongnam, Republic of Korea) for 10 min followed by a 1 min wash with 70% isopropanol. Then, cells were stained with ORO (Sigma-Aldrich, St. Louis, MO, USA) and incubated for 1 h at room temperature. Images were captured using a light microscope (Leica Microsystems; Wetzlar, Germany). After this, respective wells were destained with absolute isopropanol, and post-staining lipid quantification was performed based on absorbance at 490 nm using a microplate reader (Bio-Rad). The obtained values were presented as relative values with respect to the "mock" condition, as 1.

2.5. Determination of triglyceride contents in cells and glycerol concentrations in media

A triglyceride assay was performed using cell lysates with a TG assay kit (Triglyceride Colorimetric Assay Kit, Cayman, #10010303, Ann Arbor, MI, USA) as per the manufacturer's instructions. The glycerol assay was performed to determine the levels of glycerol released into the cell supernatant as an effect of CC on lipolysis using a glycerol assay kit (EZ-free Glycerol Assay Kit, DOGENBIO, DG-FGC100, Seoul, Republic of Korea) according to the manufacturer's guidelines.

2.6. Quantitative reverse transcriptase PCR (qRT-PCR)

Total RNA was isolated using the QIAzol Lysis Reagent (QIAGEN, Hilden, Germany), following the manufacturer's instructions. The obtained RNA was reverse transcribed to cDNA using the RevertAid™ H Minus First Strand cDNA synthesis kit (Fermentas, Hanover, NH, USA) with the following conditions: 2 U of Dnase I for 30 min at 37 °C, 50 mM EDTA for 10 min at 65 °C, a 1:1 ratio of random hexamers and Oilgo(dT) 18 primers for 5 min at 65 °C, and 10 mM of dNTP mix, 20 U of an RNase inhibitor, and 200 U of RevertAid H Minus Reverse Transcriptase for 5 min at 25 °C, 1 h at 42 °C, and 5 min at 70 °C. Then, qRT-PCR was performed using the SYBR Green I qPCR kit (Takara, Shiga, Japan). The details of the primers used are provided in Supporting Information Table S1. Mouse *Arbp* (ribosomal protein large P0) gene expression was used to normalize relative mRNA expression levels, and the fold change was calculated using the $2^{-\Delta\Delta Ct}$ method. *Garcinia cambogia* (GC) was used as a positive control wherever applied.

2.7. In vivo experiment using mice

Four-week-old C57BL/6 male mice (16.5 ± 1 g) were purchased from DBL Co., Ltd. (Chungbuk, Republic of Korea) and subjected to a 1-week acclimatization. All animal experiment procedures were carried out at Animal Care Center in Ajou University School of Medicine and approved by the Institutional Animal Care and Use Committee (2022–0005). The approval indicates compliance with ethical standards and adherence to the guidelines set by the Institutional Animal Care and Use Committee. The mice were housed in specific pathogen-free environments with an optimum temperature (23 ± 2 ℃) and day/night cycle (12/12 h) and provided with adequate chow and water. Mice were randomized based on weight and the sample size of five mice per cage in this study was justified based on considerations related to statistical analysis, ethical use of animals, and practical constraints. To minimize potential confounders, we conducted randomization, counterbalancing, blinding and standardization of procedures. The individuals conducting measurements or administering treatments are unaware of the experimental conditions. The HFD, which provides 60% of the daily calorie intake as fat, was used to induce obesity in mice. Three distinct *in vivo* experiments (80 mice in total) were performed as follows: (1) pre-administration HFD-model mice randomly allocated to five groups (25 mice in total), specifically a normal diet (ND; n = 5), HFD (n = 5), or HFD supplemented with three concentrations of CC extract (25 mg/kg/day (HFD+CC25), 50 mg/kg/day (HFD+CC50), and 100 mg/kg/day (HFD+CC100); n = 5 mice per group)); (2) post-administration, following a 4-week exclusive HFD regimen, where HFD-model mice were randomly divided into four groups (20 mice in total), specifically ND (n = 5), HFD (n = 5), or HFD supplemented with two concentrations of CC extract (50 mg/kg/day (HFD+CC50), and 100 mg/kg/day (HFD+CC100); n = 5 mice per group) for 8 weeks; and (3) post-administration, following a 6-week exclusive HFD regimen, mice were randomly divided into five groups (25 mice in total), specifically ND (n = 5), HFD (n = 5), or HFD supplemented with two concentrations of CC extract (50 mg/kg/day (HFD+CC50) and 100 mg/kg/day (HFD+CC100), with 100 mg/kg/day GC (HFD+GC100) as a positive control; n = 5 mice per group) for 6 weeks. The inclusion criteria for animals in the study were based on the observation of increased weight gain in HFD mice. Specifically, mice were included if they demonstrated a noticeable increase in weight compared to the ND group after 4 weeks of administration. In addition, mice were excluded from the study if they did not exhibit weight gain compared to the ND group under the HFD conditions. The weekly body weights of mice across the experiments were determined, and measurements from the final week were utilized to evaluate the significance of body weight reductions in the treatment groups compared to those in the control group. At the end of the experiment, mice were euthanized using CO_2, and samples of blood, epididymal adipose tissues, and livers were gathered for subsequent analyses. To evaluate the potential toxicity of CC *in vivo*, two groups (10 mice in total) (ND and ND supplemented with CC (100 mg/kg/day)) were used (n = 5 mice per group).

2.8. Oral glucose tolerance test (OGTT) and insulin tolerance test (ITT)

OGTT and ITT were performed during the final week of the experiment for the pre-treatment HFD-model group. For the OGTT, mice were fasted for 12 h, followed by the oral administration of glucose (2 g/kg; Sigma-Aldrich) via gavage. ITT was performed on mice fasted for 6 h and subsequently intraperitoneally injected with insulin (1 U/kg, Sigma-Aldrich, I0516). Blood samples were obtained from the tail vein (caudal vein), and glucose levels were monitored using a glucometer (Allmedicus, Anyang, Republic of Korea) at various time intervals (0, 20, 40, 60, 80, or 100 min). The area under the blood glucose concentration–time curve for 100 min following glucose loading or insulin injection ($AUC_{0–100\ min}$) was utilized to assess glucose uptake and insulin sensitivity.

2.9. Metabolic cage analysis

A metabolic cage (TSE PhenoMaster, TSE Systems, Bad Homburg, Germany) accommodating 12 mice was utilized. During the final week of the experiment, the analysis was conducted specifically for the pre-treatment HFD-model group to evaluate the metabolic profiles. Each mouse from the ND, HFD, and two CC-administered groups (50 and 100 mg/kg/day) was included in the analysis. Individual measurements were taken for food and water consumption. Additionally, assessments included the measurement of oxygen consumption (VO_2), CO_2 expiration (VCO_2), and mobility. These measurements facilitated the calculation of the respiratory exchange ratio (RER) and energy expenditure (EE). Moreover, the average total body percentage composition of fat and lean body mass was determined as part of the analysis.

2.10. Blood plasma biochemical analysis

Blood samples were obtained via cardiac puncture and centrifuged at 2,500 rpm for 15 min at 4 °C; the obtained plasma was stored at −80 °C and later used to analyze the levels of high-density lipoprotein (HDL), low-density lipoprotein (LDL), triglycerides, total cholesterol, and liver enzymes, namely aspartate aminotransferase (AST) and alanine aminotransferase (ALT), using a dry chemistry analyzer (DRI-CHEM NX500, Fujifilm Global, Minato-ku, Tokyo, Japan).

2.11. Histological analysis of the liver and epididymal adipose tissues and quantification

Liver and epididymal adipose tissues were fixed in 4% paraformaldehyde, sectioned (3 μm; Leica Microsystems, Wetzler, Germany) with a microtome, and stained with hematoxylin and eosin (H&E; SSN Solutions, London, UK). Stained images were scanned using an Axioscan Z1 slide scanner (Carl Zeiss). ImageJ software was used to quantify the number of lipid vacuoles in the liver and the size of the adipocytes in adipose tissue.

2.12. Statistical analysis

Statistical analysis was performed using GraphPad Prism 9.0 (GraphPad, San Diego, CA, USA), and the results were presented as the mean ± standard error of the mean. Two groups were compared using an unpaired t-test, and multiple comparisons were performed using ANOVA followed by Tukey's honestly significant difference post hoc test, unless otherwise mentioned. A "*p*" value less than 0.05 was considered statistically significant.

3. Results

3.1. Anti-adipogenic effects of CC in vitro

We first investigated the anti-adipogenic effect of CC using mouse 3T3-L1 pre-adipocyte cells. Cells were induced with MDI to promote adipocyte differentiation and then co-incubated with either three concentrations of CC (10, 20, and 50 μg/mL) or GC (50 μg/mL), used as a positive control. Lipid droplet accumulation in differentiated cells was assessed using ORO. Additionally, we quantified the mRNA expression levels of key adipogenesis factors, including *Pparg*, *Cebpa*, *Fabp4*, and *Fasn*, along with lipolytic factors, including *Hsl*, *Acox1*, and *Cpt2*, via qRT-PCR. Further analyses included measurements of intracellular triglyceride (TG) contents and glycerol concentrations in the cell supernatant. Results revealed that adipocyte differentiation induced by MDI significantly increased lipid accumulation and concurrently upregulated the expression of adipogenesis-related factors, with reduced TG contents compared to levels in the control group. However, CC treatment effectively alleviated lipid droplet accumulation in differentiated cells (Fig. 1A-B). We also observed a dose-dependent reduction in the mRNA expression levels of adipogenesis-related factors (*Pparg*, *Cebpa*, *Fabp4*, and *Fasn*) (Fig. 1C) accompanied by an increase in lipolysis-related factors (*Hsl*, *Acox1*, and *Cpt2*) (Fig. 1D). Moreover, CC treatment significantly decreased intracellular TG contents and increased glycerol concentrations, compared to those in the untreated mock group

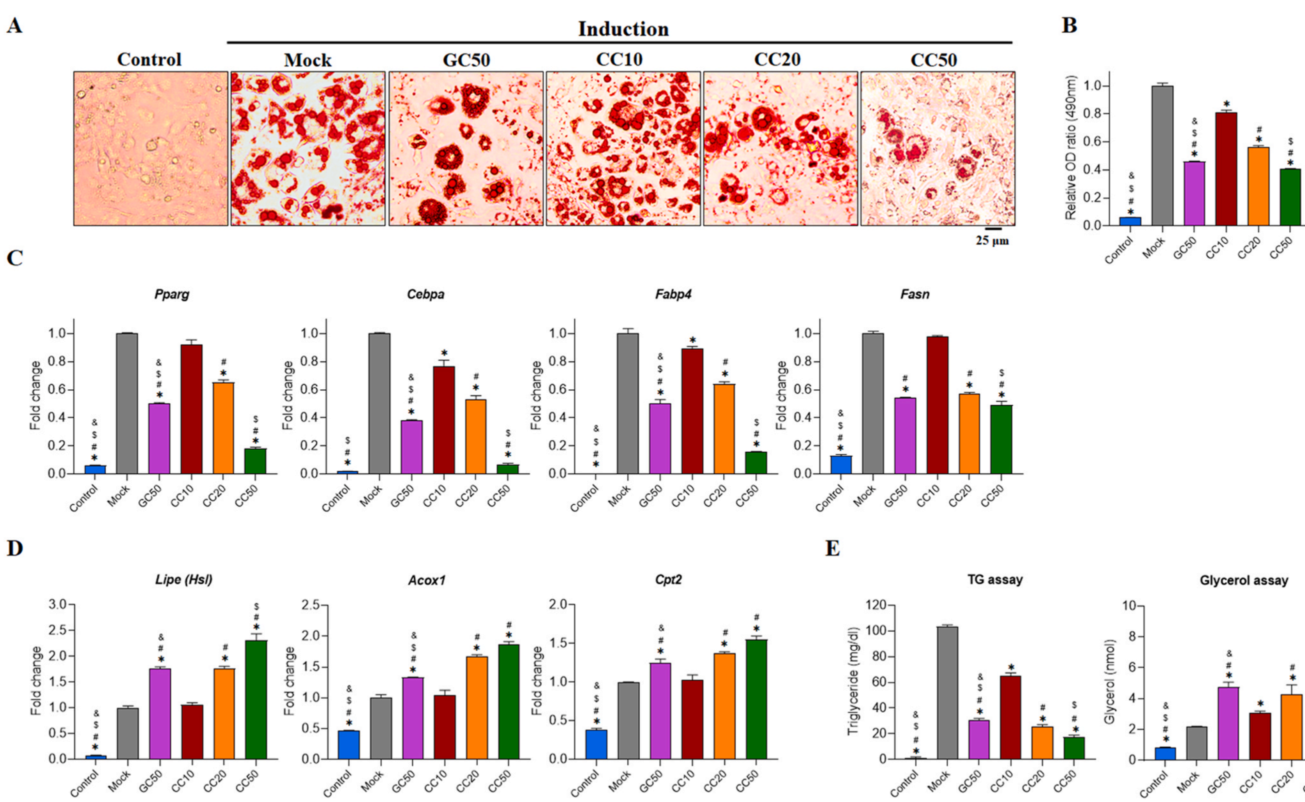

Fig. 1. Inhibitory effect of *Celosia cristata* (CC) on mouse 3T3-L1 pre-adipocyte cell differentiation. Cells were induced to differentiate into adipocytes through treatment with MDI (3-isobutyl-1-methylxanthine, dexamethasone, and insulin) for 3 days, followed by 5 days of insulin treatment. During the 8 days of the experiment, varying concentrations (10, 20, and 50 μg/mL) of CC or 50 μg/mL *Garcinia cambogia* (GC) were administered. (**A** and **B**) Oil-red O-positive cells were visualized under a light microscope, and stained cells were quantified based on the relative absorbance at 490 nm using a microplate reader. (**C**) mRNA expression levels of adipogenic (*Pparg*, *Cebpa*, *Fabp4*, *Fasn*) and (**D**) lipolytic factors (*Hsl*, *Acox1*, *Cpt2*) were analyzed via qRT-PCR. (**E**) Levels of triglycerides (TGs; left) and glycerol (right) were determined by performing a TG and glycerol assay. All experiments were performed at least in triplicate independent experiments with duplicate samples. Statistical differences among multiple groups were assessed using a one-way analysis of variance (ANOVA), followed by post hoc analysis was performed using Tukey's honestly significant difference test. $^{*}p < 0.05$ vs. Mock, $^{\#}p < 0.05$ vs. CC10, $^{\$}p < 0.05$ vs. CC20, $^{\&}p < 0.05$ vs. CC50.

(Fig. 1E).

We conducted additional investigations to confirm the anti-adipogenic effect of CC on mouse primary-cultured ADSCs isolated from mouse adipose tissues. For the differentiation of ADSCs, cells were treated with Mesencult™ differentiation medium and co-cultured with either CC (10, 20, and 50 μg/mL) or GC (50 μg/mL) as a positive control. As expected, CC treatment decreased the number of ORO-positive cells and reduced the mRNA expression levels of key adipogenesis-related factors, including *Pparg*, *Cebpa*, *Fabp4*, and *Fasn* (Fig. 2A-C). Conversely, CC promoted an increase in the levels of lipolysis-related factors (Fig. 2D). In addition, CC administration significantly reduced the intracellular TG content and increased the glycerol concentration as compared to those in the untreated mock group (Fig. 2E). Further, CC did not have a significant effect on cellular viability in 3T3-L1 cells and ADSCs (Supporting Information Fig. S1-S2), indicating that it has no cytotoxic effects. These results suggest that the CC inhibits adipocyte differentiation and adipogenesis by downregulating the expression of adipogenesis-related factors and upregulating lipolysis-related factors in both mouse 3T3-L1 pre-adipocytes and ADSCs.

The CC extract was analyzed by HPLC and LC-MS/MS to determine marker compound for quality control. In our extraction condition, kaempferol glycosides is the main compound of the CC extract (Supporting Information Fig. S3 and S4, and Table S2). Several small peaks were also detected in the CC extract through HPLC and LC-MS/MS analysis. However, due to their minor amounts, identifying these minor compounds proved challenging. It is hypothesized that these minor compounds may contribute to the diverse bioactivity of the CC extract and potentially have synergistic effects with kaempferol, thereby enhancing the anti-obesity effects of the extract. Additionally, HPLC finger-printing analysis of the CC extract showed the predominant presence of KMP, as confirmed based on a comparison with a commercially available standard compound (Chengdu Biopurify Phytochemicals Ltd., Sichuan, China) (Supporting Information Fig. S5A-B). Flavonol glycosides were also detected in the CC extract using HPLC. However, owing to the limited commercial availability of reference compounds for flavonol glycosides, the analytical determination of individual flavonol glycosides in plant extracts proves challenging. Therefore, flavonol glycosides are normally acid-hydrolyzed, breaking them down into their aglycone forms for identification and quantification [37,38]. In this study, the CC extract was hydrolyzed following the methodology outlined in the Health Functional Food Code of the Ministry of Food and Drug Safety. The major flavonol aglycones were identified and quantified via HPLC after acid hydrolysis. The primary flavonol glycosides within the CC extract were expected to be the derivatives of KMP linked with glucose and rhamnose in different linkage forms and numbers. The KMP glycoside content was determined after acid hydrolysis to convert the glycosides to the corresponding KMP. Subsequently, the bioactive component, KMP, was used to determine the anti-adipogenic effect *in vitro* using 3T3-L1 cells. We observed that KMP treatment decreased the number of ORO-positive cells and suppressed mRNA expression (*Pparg* and *Cebpa)* compared to those in the control group (Supporting Information Fig. S5C-E). These results suggest that KMP inhibits lipid droplet accumulation via downregulation of the expression of adipogenic factors. These findings align with those of a previous study on the effects of KMP isolated from different plant sources, similarly observed using 3T3-L1 cells [36].

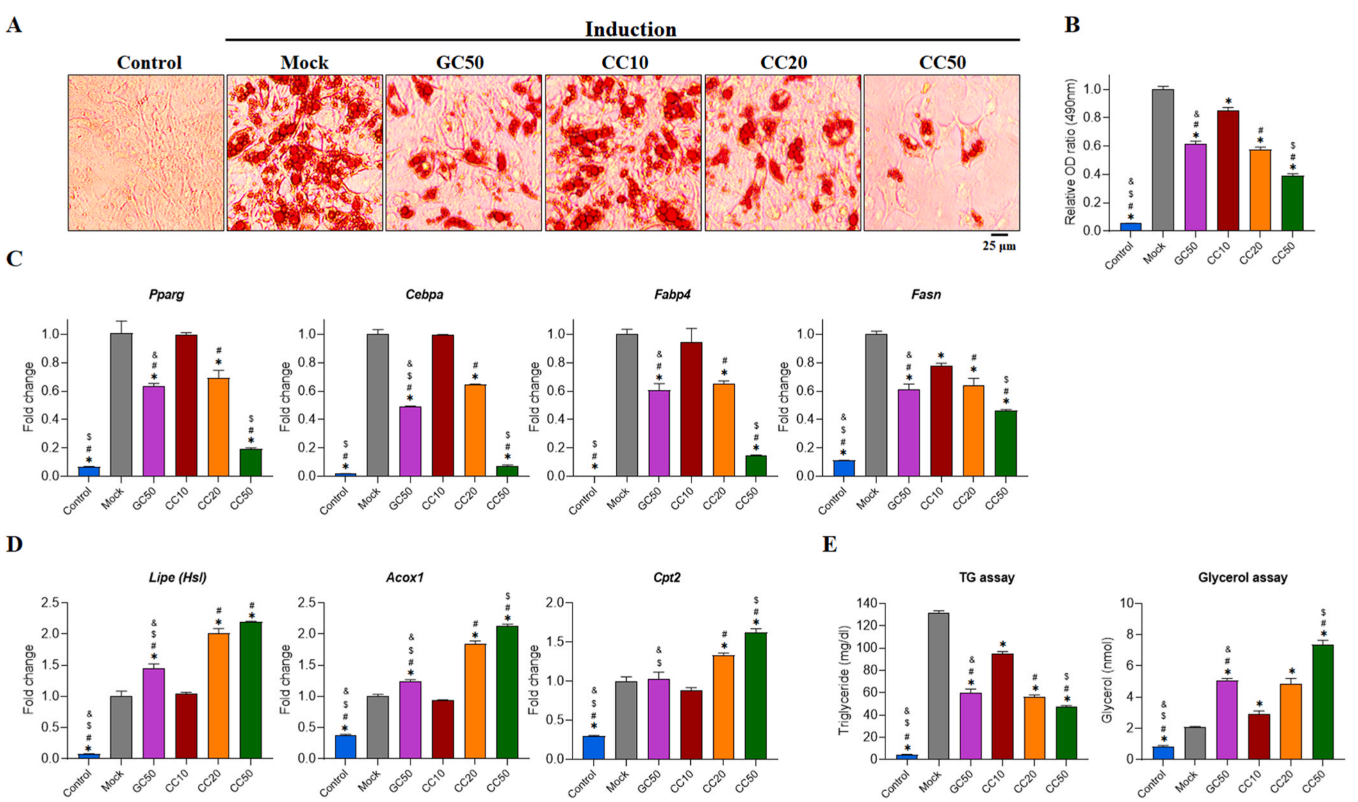

Fig. 2. Inhibitory effect of *Celosia cristata* (CC) on mouse primary adipose-derived stem cell (ADSC) differentiation. ADSCs were induced to undergo adipocyte differentiation and co-treated with varying concentrations (10, 20, and 50 μg/mL) of CC or 50 μg/mL *Garcinia cambogia* (GC). (**A** and **B**) Oil-red O-positive cells were visualized under a light microscope, and stained cells were quantified based on the relative absorbance at 490 nm using a microplate reader. (**C**) mRNA expression levels of adipogenic (*Pparg*, *Cebpa*, *Fabp4*, *Fasn*) and (**D**) lipolytic factors (*Hsl*, *Acox1*, *Cpt2*) were determined via qRT-PCR. (**E**) Levels of triglycerides (TGs; left) and glycerol (right) were determined by performing a TG and glycerol assay. All experiments were performed at least in triplicate with duplicate samples. Statistical differences among multiple groups were determined by two-way ANOVA comparisons. $^{*}p < 0.05$ vs. Mock, $^{\#}p < 0.05$ vs. CC10, $^{\$}p < 0.05$ vs. CC20, $^{\&}p < 0.05$ vs. CC50.

3.2. Effect of CC on obesity in HFD-fed mice

Based on the *in vitro* results, we investigated the potential ameliorative effects of CC on obesity using an HFD-induced mouse model. To evaluate the *in vivo* anti-obesity efficacy of CC, we conducted three different animal experiments over 12 weeks. These experiments included (1) a pre-administration phase involving 12-week HFD feeding, (2) a post-administration phase involving 4 weeks of an HFD followed by an additional 8 weeks of an HFD supplemented with CC, and (3) a post-administration phase involving 6 weeks of an HFD followed by 6 weeks of an ND supplemented with CC. The pre-treatment experiment was designed to evaluate the effects of CC on preventing obesity in mice exposed to a 12-week HFD. Subsequently, the post-treatment HFD experiment was aimed at investigating the effects of CC on mitigating weight gain in obese mice over 4 weeks of HFD feeding. Additionally, the post-treatment ND experiment was conducted to assess the efficacy of CC in managing weight loss by transitioning mice with obesity-induced for 6 weeks via HFD feeding to a normal diet supplemented with CC for the following 6 weeks. These distinct diet patterns were designed considering varying degrees of obesity and the potential dietary modifications during the weight-loss process. The approach accounted for different starting points in weight-management strategies and allowed for adjustments in the diet status during the weight-loss period.

To establish the *in vivo* toxicity profile of CC, mice were administered either an ND or ND supplemented with 100 mg/kg/day of CC for 12 weeks. Results showed no significant alterations in body weights (Supporting Information Fig. S6A-B) and food intakes (Supporting Information Fig. S6C), organ functions, morbidity, mortality, clinical manifestations, or observed abnormalities, such as alterations in behavior, instances of diarrhea, or variations in water intake. This indicated that CC had no toxic effects *in vivo*.

3.2.1. Effects of CC on preventing obesity in HFD-fed mice

In the pre-treatment experiment, mice were administered an ND, HFD, or HFD supplemented with three concentrations (25, 50, or 100 mg/kg/day) of CC for 12 weeks, with weekly body weight measurements. At the end of the experiment, the histomorphological analysis results based on the liver and adipose tissues were visualized via H&E staining, and the mRNA expression levels of adipogenic (*Pparg*, *Cebpa*, *Fabp4*, and *Fasn*) and lipolytic factors (*Hsl*, *Acox1*, and *Cpt2*) in adipose tissue were analyzed via qRT-PCR. Blood plasma samples were collected to analyze parameters including HDL, LDL, triglycerides, total cholesterol, and liver enzymes (AST and ALT). Glucose metabolism was evaluated via OGTT and ITT.

Results showed a significant difference in body weights between

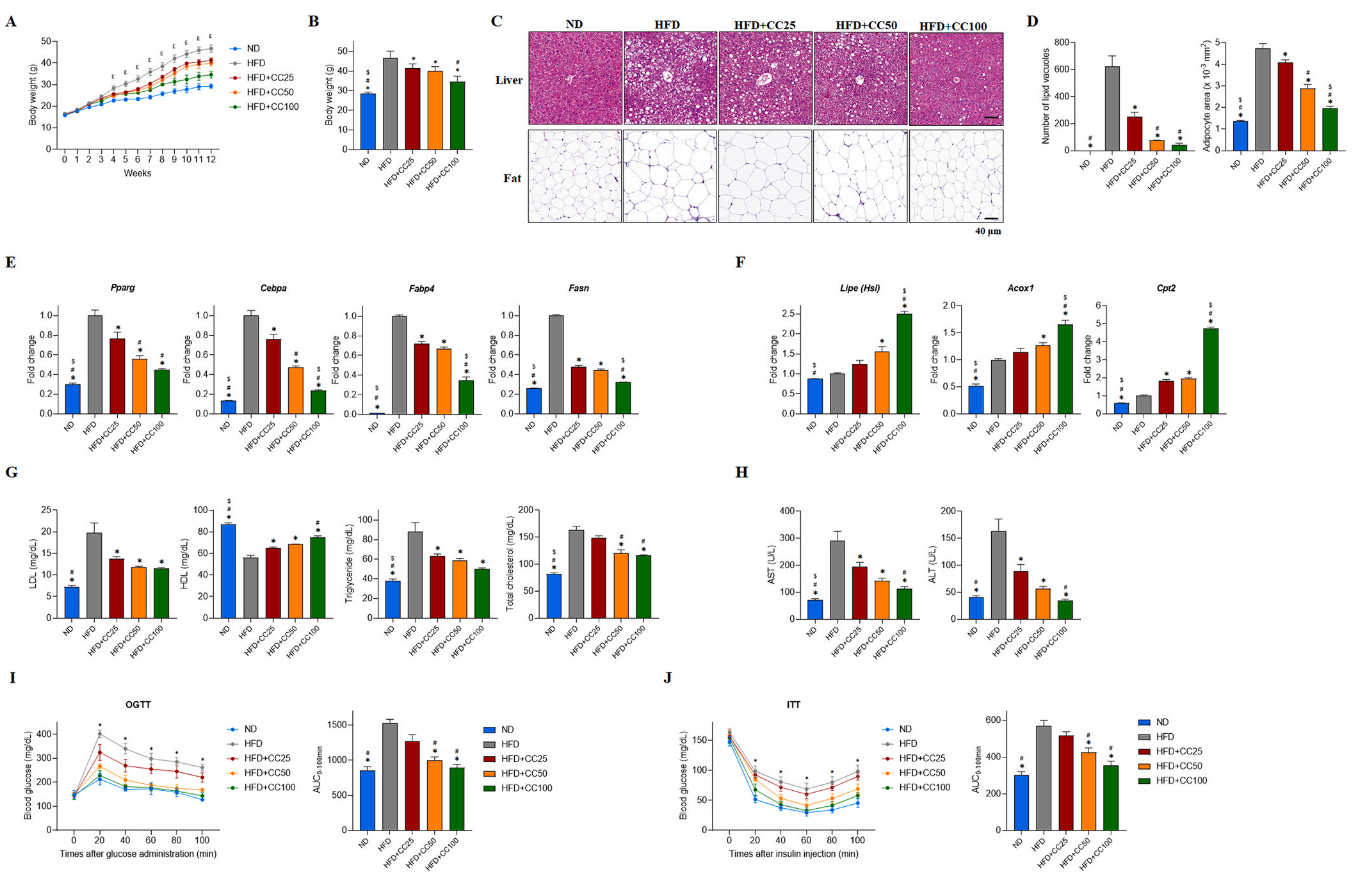

Fig. 3. Ameliorative effects of pre-administered *Celosia cristata* (CC) on body weight gain and various obesity-related parameters in HFD-fed mice. Mice were provided with either an ND or HFD supplemented with a CC total extract at three concentrations (25, 50, and 100 mg/kg/day; n = 5 mice per group) for 12 weeks. Weekly body weight changes were monitored, and the liver, adipose tissue, and blood were analyzed to determine the effect of CC on obesity and related factors. (**A**) Weekly changes in body weights over the 12-week experimental period. (**B**) Total mouse body weights at the end of the experiment. (**C**) Hematoxylin and eosin (H&E) staining of mouse liver and adipose tissue. (**D**) Quantification of lipid vacuoles in the liver (left) and adipose tissue area (right), using ImageJ software. (**E**) mRNA expression levels of adipogenic (*Pparg*, *Cebpa*, *Fabp4*, *Fasn*) and (**F**) lipolytic factors (*Hsl*, *Acox1*, *Cpt2*), determined via qRT-PCR. (**G**) Blood plasma analysis of lipid profiles (LDL, low-density lipoprotein; HDL, high-density lipoprotein; triglycerides; and total cholesterol) and (**H**) evaluation of liver enzymes (AST, aspartate aminotransferase; ALT, alanine aminotransferase) using a DRI-CHEM NX500. (**I**) Time course of blood glucose levels during the (left) oral glucose tolerance test (OGTT) and (**J**) during the total insulin tolerance test (ITT), with respective calculated $AUC_{0-100\ min}$ values (right). Statistical differences of multiple groups were analyzed by ANOVA, followed by post hoc analysis was performed using Tukey's honestly significant difference test. ND, normal diet; HFD, high-fat diet. $^{*}p < 0.05$ vs. HFD, $^{\#}p < 0.05$ vs· HFD+CC25, $^{\$}p < 0.05$ vs. HFD+CC50· $^{£}p < 0.05$ HFD vs. HFD+CC100.

HFD- and CC-treated HFD groups, becoming apparent after the initial 4 weeks of the experiment (Fig. 3A). Throughout the 12 weeks, mice in all CC-treated groups exhibited significant reductions in body weights (Fig. 3A-B) without changing daily food intakes (Supporting Information Fig. S7). Moreover, liver and adipose tissue analysis of HFD-treated mice revealed excessive intrahepatic lipid accumulation and adipocyte hypertrophy compared to those in the ND group (Fig. 3C-D). However, CC treatment mitigated fat vacuole accumulation in the liver and reduced adipocyte expansion in the adipose tissue (Fig. 3C-D). Additionally, CC administration downregulated the expression of adipogenic factors (Fig. 3E) while upregulating lipolytic factors in a dose-dependent manner in HFD-fed mice (Fig. 3F). CC supplementation also improved lipid profile parameters, increasing HDL while decreasing LDL, triglycerides, and total cholesterol (Fig. 3G) and reduced liver enzymes such as AST and ALT as compared to levels in the HFD group (Fig. 3H). In addition, CC significantly enhanced glucose metabolism based on OGTT (Fig. 3I) and ITT (Fig. 3J), as compared to that in the HFD group, indicating that it could promote glucose utilization and insulin sensitivity. Significantly noticeable changes were observed in groups supplemented with 50 and 100 mg/kg/day of CC compared to observations in the HFD group. Consequently, these concentrations of CC were selected for subsequent *in vivo* studies.

In the final week of the experiment, the effects of CC on systemic metabolism were investigated using a metabolic cage monitoring system and the mRNA expression levels of mitochondrial energy metabolism related markers (adipocyte browning, mitochondrial biogenesis and β-oxidation) were analyzed by qRT-PCR. We included mice from groups fed an ND, HFD, and HFD supplemented with two concentrations of CC yielding statistically significant results (50, 100 mg/kg/day), focusing on the inhibition of body weight gains. Metabolic parameters, including VO_2, VCO_2, RER, and EE, were recorded for both dark and light phases over a 48-h period (Supporting Information Fig. S8A-B). Result revealed a lack of statistically significant changes in VO2, VCO2, RER, and EE in mice during different light and dark phases This phenomenon is attributable to the metabolic adaptation of mice to the light-dark cycle, resulting in minimal variations in metabolic parameters across different phases as previously described [39]. Subsequently, the effects of CC on the average total body fat and lean body mass composition were calculated. The mice fed an HFD supplemented with CC exhibited a decrease in their average body fat percentage and an increase in their lean body mass composition, compared to those in the HFD group (Fig. 4A). These enhancements in body composition were accompanied by improvements in metabolic parameters, including an increased average EE, RER, VO_2, and VCO_2 (Fig. 4B-C and Supporting Information Fig. S8A-B). CC supplementation thus inhibited body weight gain by improving obesity-related parameters and metabolic profiles in mice subjected to 12 weeks of HFD administration. After observing enhanced metabolic profiles in CC administered HFD-induced mice during metabolic cage monitoring, we conducted an additional experiment to analyze the mRNA expression levels of adipocyte browning (*Ucp1*, *Prdm16*, and *Pgc1a*), mitochondrial biogenesis (*Cytc*, and *Cox4b*) and β-oxidation (*Mccad*) associated markers in mice adipose tissue. Results showed that significant upregulations of adipocyte browning, mitochondrial biogenesis and β-oxidation markers were observed in CC administered group as compared to HFD group (Fig. 4D-F). This finding supports the observed enhanced metabolic profiles and the subsequent inhibition of body weight gain in mice, which may be attributed to increased mitochondrial activities resulting from the upregulation of adipocyte browning, mitochondrial biogenesis, and β-oxidation associated markers.

3.2.2. *Effects of CC on mitigating weight gain in established HFD-induced obese mice*

To investigate the effects of CC on mice with established obesity, we started mice on an HFD to induce weight gain and divided them into two post-administration groups. One group received an HFD for 4 weeks to establish obesity, followed by 8 weeks of continuous CC administration while maintaining the HFD. The other group received an HFD for 6 weeks to increase obesity levels, and then the effects of CC were assessed after switching the diet from HFD to ND supplemented with CC for 6 weeks. Additionally, a well-known anti-obesity supplement, GC (100 mg/kg/day), was included as a positive control in the third *in vivo* experiment for comparison. Throughout these experiments, we monitored weekly body weight changes, performed histomorphological analyses of liver and epididymal fat tissues, assessed the mRNA expression of adipogenic (*Pparg*, *Cebpa*, *Fabp4*, and *Fasn*) and lipolytic factors (*Hsl*, *Acox1*, and *Cpt2*), and analyzed plasma lipid profiles (LDL, HDL, triglycerides, and total cholesterol) and liver enzymes (AST and ALT).

Since obesity was successfully induced by the HFD at 4 weeks in the post-treatment HFD experiment, the HFD was supplemented with CC extract (50 and 100 mg/kg/day) from 5 weeks of HFD feeding (Fig. 5A). Subsequent CC treatment (50 and 100 mg/kg/day) significantly reduced the total body weight gain in HFD-fed mice (Fig. 5B) without altering daily food intakes (Supporting Information Fig. S9). Further, histological assessments revealed decreased hepatic lipid accumulation in the liver and adipocyte enlargement in epididymal fat tissues (Fig. 5C-D). Moreover, levels of adipogenic factors were decreased, whereas those of lipolytic factors were increased, at the mRNA level in the adipose tissues of CC-treated mice (Fig. 5E-F). The administration of CC resulted in a significant increase in HDL levels and a decrease in LDL, triglyceride, and total cholesterol levels (Fig. 5G), along with a decrease in AST and ALT levels (Fig. 5H), as compared to those in the HFD group. These results indicate that CC inhibits body weight gain, with improvements in obesity-related parameters, by regulating genes related to adipogenesis and lipolysis under post-administration HFD conditions.

In the post-treatment ND experiment, mice were fed an HFD for 6 weeks for the induction of obesity, with CC administered, and the diet was switched from an HFD to ND for 6 weeks, based on an evaluation of the effects of CC under ND conditions in obese mice. Moreover, the anti-obesity supplement GC (100 mg/kg/day) was used as a positive control. After 6 weeks of HFD feeding, total body weights were reduced in all mice during the diet-switching (ND) period. Further, the effects of CC on inhibiting body weight gains became apparent from the 8th week (2 weeks after CC treatment) of the experiment, with a significant reduction observed with CC and GC doses of 100 mg/kg/day (Fig. 6A-B), while maintaining consistent daily food intakes (Supporting Information Fig. S10). Histological analyses indicated a reduction in hepatic lipid vacuole accumulation and adipocyte hypertrophy in the liver and adipose tissues, respectively (Fig. 6C-D). Additionally, levels of adipogenic factors decreased, whereas those of lipolytic factors increased, at the mRNA level, among adipose factors, as compared to those in the group switching from an HFD to an ND (Fig. 6E-F). Moreover, HDL levels were increased, whereas LDL, triglyceride, and total cholesterol (Fig. 6G) and AST and ALT levels were decreased, compared to those in the group who switched from an HFD to an ND (Fig. 6H). Collectively, the findings from three distinct *in vivo* experiments indicate the potential effects of CC on weight management, liver and adipose tissue morphology, gene expression related to adipogenesis and lipolysis, lipid profiles, liver enzyme activities, and glucose metabolism in an HFD-induced obese mouse model.

4. Discussion

Obesity is a widespread public health issue frequently associated with metabolic disorders, with an increasing prevalence over time, prompting the development of diverse therapeutic interventions [1,40]. Although various synthetic drugs have been formulated to address obesity, they impose a risk of significant side effects and safety concerns on patients with long-term usage [23]. As a result, plant-based natural supplements have been explored as potential alternatives owing to their lower toxicity and reduced side effects compared to synthetic medications [24]. This study aimed to determine the potential anti-obesity

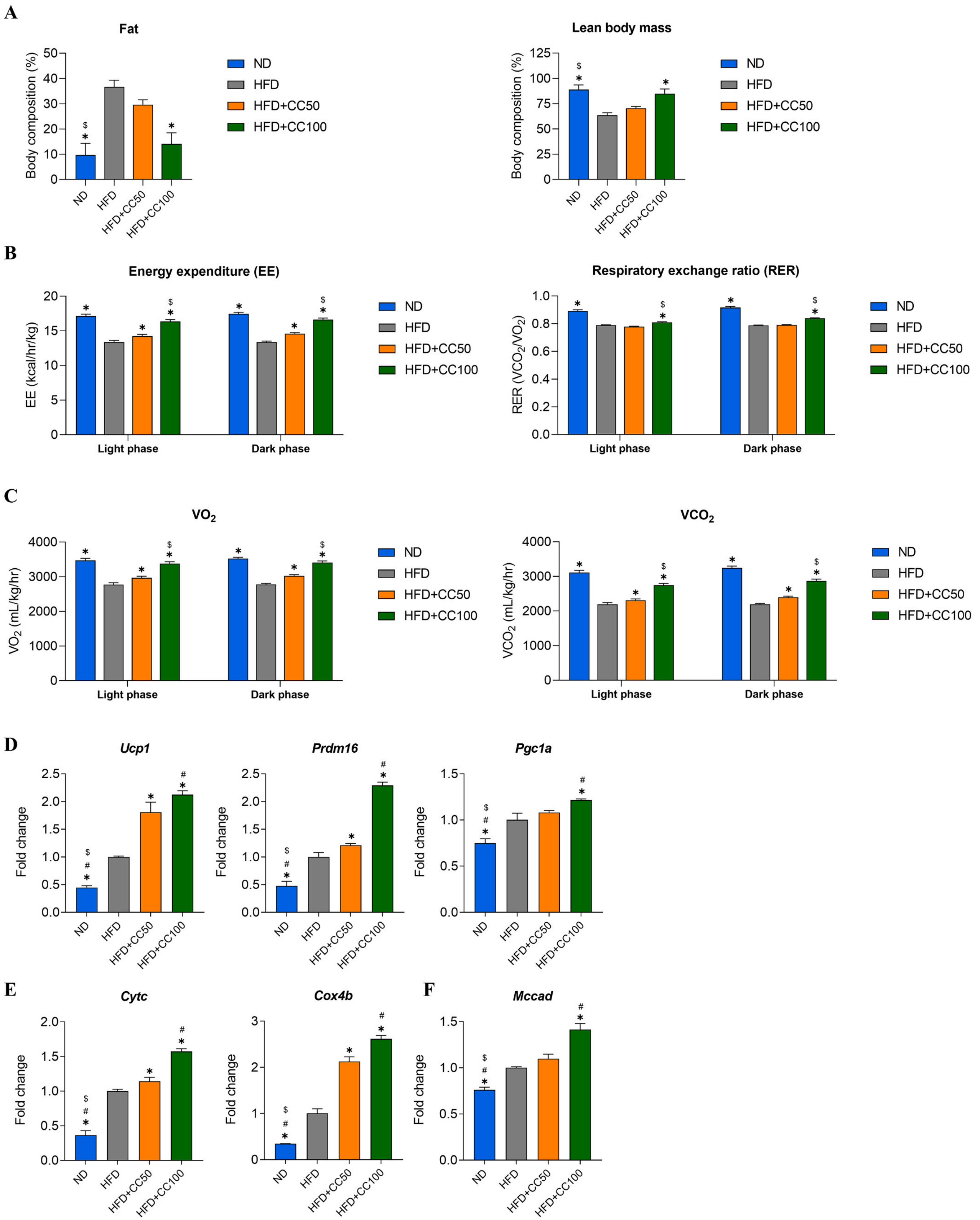

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Fig. 4. Preventive effects of pre-administered *Celosia cristata* (CC) on the metabolic profiles and mRNA expression levels of mitochondrial energy metabolism related markers on HFD-fed mice. Mice were divided into four groups (n = 3 mice per group) receiving an ND, HFD, or HFD supplemented with the two most effective concentrations of CC (50 and 100 mg/kg/day) during the pre-administration phase. Metabolic profiles were analyzed during the final week of the 12-week experiment. Metabolic profiles including (**A**) average body fat composition (left) and lean body mass (right), (**B**) average energy expenditure (EE) (left) and respiratory exchange ratio (ERE) (right), and (**C**) average oxygen (VO_2) consumption (left) and average carbon dioxide (VCO_2) production (right) were measured. (**D**) mRNA expression levels of adipocytes browning associated markers (*Ucp1, Prdm16, Pgc1a*). (**E**) mRNA expression levels of mitochondrial biogenesis associated markers (*Cytc, Cox4b*). (**F**) mRNA expression levels of mitochondrial β-oxidation related marker (*Mccad*). The results presented on the bar graph are from at least three dependent experiments. Statistical differences of multiple groups in metabolic cage parameters were analyzed by two-way ANOVA multiple comparisons. ND, normal diet; HFD, high-fat diet. $^*p < 0.05$ vs. HFD, $^{\$}p < 0.05$ vs HFD+CC50. For mRNA expression levels statistical analysis among the groups were assessed using a one-way analysis of variance (ANOVA), followed by post-hoc analysis using Tukey's honestly significant difference test. $^*p < 0.05$ vs. HFD, $^{\#}p < 0.05$ vs. HFD+CC50, $^{\$}p < 0.05$ vs. HFD+CC100.

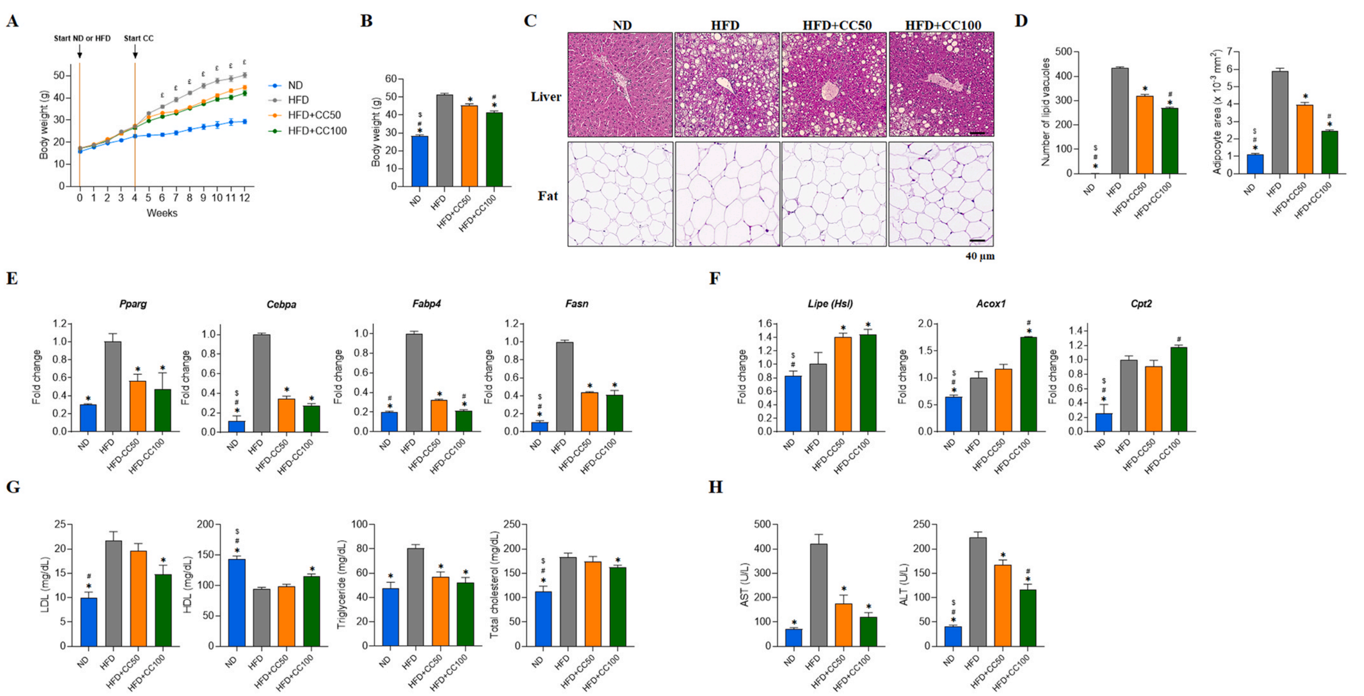

Fig. 5. Anti-obesity effects of post-administered *Celosia cristata* (CC) on HFD-fed mice. Mice were provided with either an ND or HFD supplemented with a CC total extract at two concentrations (50, and 100 mg/kg/day; n = 5 mice per group) for 12 weeks. (**A**) Body weight changes in mice during the 12-week experiment. (**B**) Total mouse body weights at the end of the experiment. (**C**) H&E staining of mouse liver and adipose tissue. (**D**) Quantification of lipid vacuoles in the liver (left) and adipose tissue areas (right) using ImageJ software. (**E**) mRNA expression levels of adipogenic (*Pparg*, *Cebpa*, *Fabp4*, *Fasn*) and (**F**) lipolytic factors (*Hsl*, *Acox1*, *Cpt2*), determined via qRT-PCR. (**G**) Blood plasma analysis of lipids (LDL, low-density lipoprotein; HDL, high-density lipoprotein; triglycerides; and total cholesterol) and (**H**) evaluation of liver enzymes (AST, aspartate aminotransferase; ALT, alanine aminotransferase), using a DRI-CHEM NX500. The results presented on the bar graph are from at least three dependent experiments. Statistical differences of multiple groups were analyzed by two-way ANOVA multiple comparisons. ND, normal diet; HFD, high-fat diet. $^*p < 0.05$ vs. HFD, $^{\#}p < 0.05$ vs HFD+50, $^{\$}p < 0.05$ vs. HFD+ CC100, $^{£}p < 0.05$ HFD vs. HFD+CC100.

effects of CC using *in vitro* models of mouse 3T3-L1 pre-adipocytes and primary ADSCs and an *in vivo* model of 60% HFD-induced obese mice, evaluating various obesity-related parameters.

Understanding the processes governing adipogenesis and lipolysis has an important role in strategies aimed at regulating adipocyte differentiation and managing obesity-related disorders [41]. Obesity primarily manifests as the excessive accumulation of fat, leading to increased adipogenesis, during which energy is stored in the form of lipids/TGs while lipolytic activities are delayed [7,9]. The rate of fat accumulation is primarily balanced by the interplay between adipogenesis and lipolysis [42]. Key transcription factors significantly involved in enhancing adipocyte differentiation include Pparg, Cebpa, Fabp4, and Fasn [10]. Pparg functions as a central regulator of adipogenesis, crucial for maintaining adipocyte differentiation and insulin sensitivity [43]. Cebpa, upon binding to *Pparg* promoters, induces the expression of *Pparg* isoform 2, thereby intensifying adipogenesis [44]. Meanwhile, Fasn facilitates the production of long-chain fatty acids, promotes lipogenesis, and is associated with metabolic alterations during overnutrition [45]. Fabp4, expressed in differentiated adipocytes, promotes lipogenesis by supporting lipid uptake [46]. Both Fabp4 and Fasn play pivotal roles as lipogenic proteins and enzymes involved in lipogenesis [47]. Conversely, lipolysis factors, such as Hsl, Acox1, and Cpt2, promote lipid breakdown to generate energy [11–13]. Hsl is a neutral intracellular lipase that is highly expressed in adipose tissue, promoting the hydrolysis of triacylglycerol [11]. Further, Acox1 is an essential enzyme that initiates the peroxisomal fatty acid β-oxidation pathway and is critical for maintaining lipid homeostasis [12]. Cpt2, an enzyme stimulating the beta-oxidation of long-chain mitochondrial fatty acids to acetyl-coenzyme A, plays a crucial role in the synthesis of fatty acids, cholesterol, and triglycerides [13,48]. Glycerol release by adipose tissue serves as a prominent indicator of TG hydrolysis [49], and quantifying triglyceride and glycerol contents can be used to indicate the metabolic status of cells undergoing adipogenic differentiation [50]. In this study, CC treatment inhibited lipid droplet accumulation during adipocyte differentiation in 3T3-L1 cells and ADSCs. This inhibition is mediated by the downregulation of key adipogenic transcription factors such as *Pparg*, *Cebpa*, *Fabp4*, and *Fasn* and upregulation of lipolysis related factor including *Hsl*, *Acox1*, and *Cpt2* both *in vitro* (in 3T3-L1

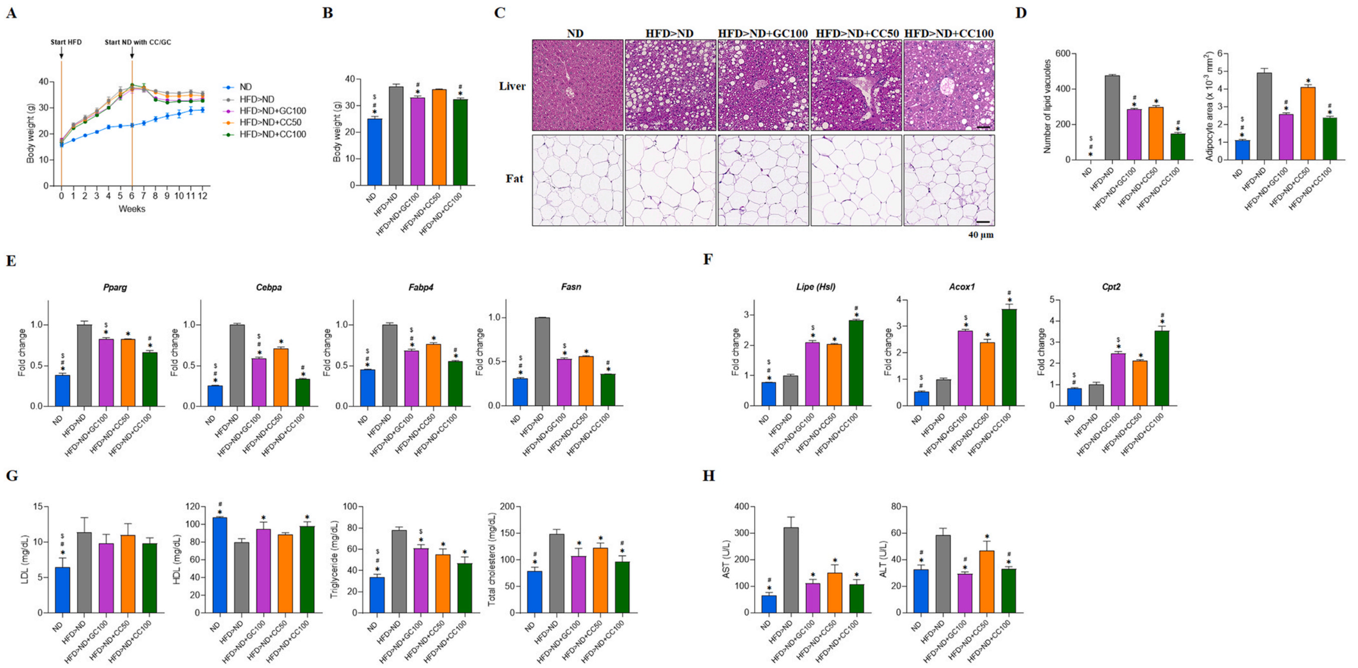

Fig. 6. Post-treatment effect of *Celosia cristata* (CC) with an ND on body weight gain in HFD-treated mice with established obesity. Mice were provided with either an ND or HFD supplemented with a CC total extract at two concentrations (50, and 100 mg/kg/day) or GC (100 mg/kg/day) for 12 weeks (n = 5 mice per group). (**A**) Body weight changes in mice during the 12 weeks. (**B**) Total mouse body weights at the end of the experiment. (**C**) Hematoxylin and eosin (H&E) staining of mouse liver and adipose tissue. (**D**) Quantification of lipid vacuoles in the liver (left) and adipose tissue area (right) using ImageJ software. (**E**) mRNA expression levels of adipogenic (*Pparg*, *Cebpa*, *Fabp4*, *Fasn*) and (**F**) lipolytic factors (*Hsl*, *Acox1*, *Cpt2*), determined via qRT-PCR. (**G**) Blood plasma analysis of lipid profiles (LDL, low-density lipoprotein; HDL, high-density lipoprotein; triglycerides; and total cholesterol) and (**H**) liver enzymes (AST, aspartate aminotransferase; ALT, alanine aminotransferase) analyzed using a DRI-CHEM NX500. ND, normal diet; HFD, high-fat diet. The results presented on the bar graph are from at least three dependent experiments. Statistical differences of multiple groups were analyzed by two-way ANOVA multiple comparisons. ${}^{*}p < 0.05$ vs. HFD>ND, ${}^{\#}p < 0.05$ vs. HFD>ND+CC50, ${}^{\$}p < 0.05$ vs. HFD>ND+CC100, ${}^{£}p < 0.05$ HFD>ND vs. HFD>ND+CC100.

pre-adipocytes and ADSCs) and *in vivo* (in HFD-induced obese mice). The observed anti-adipogenic effects of CC were supported by the decreased triglyceride concentration and increased glycerol release. Our results suggested that CC treatment prevents adipogenesis and fat accumulation and exerts a modulatory effect on reducing adipogenesis-related factors and increasing lipolysis-related factors during adipocyte differentiation.

An HFD can be used to induce a dependable dietary model of obesity in mice, which is commonly employed to assess anti-obesity effects *in vivo* [51]. The increased intake of calorie-dense food aggravates the accumulation of excessive fat *in vivo*, effectively promoting obesity [52]. Persistent undesirable weight gain is closely associated with metabolic peculiarities, including hepatic steatosis, enlarged adipocytes, hyperlipidemia, and disrupted blood glucose regulation [1]. Hepatic steatosis and adipocyte hypertrophy are hallmark features of obesity that occur due to the excessive deposition of fat [14]. Hepatic steatosis, resulting from an overabundance of fat, can trigger elevated levels of liver enzymes, such as plasma ALT and AST [53]. In turn, AST and ALT are reliable markers used to measure liver functions, and they are released after hepatocellular damage [54]. One study previously showed the beneficial effects of CC on protecting against hepatic steatosis, non-alcoholic fatty liver disease, and oxidative stress in mice [28]. In this study, CC was found to have an ameliorative effect on severe hepatic steatosis and adipocyte hypertrophy, concurrently leading to a reduction in plasma ALT and AST levels in HFD-induced obese mice. These findings suggest that CC can effectively mitigate fat accumulation in the liver and adipose tissues, consequently reducing damage-associated liver enzymes.

Obesity leads to a high risk of type 2 diabetes due to disrupted glucose metabolism and insulin resistance [55]. Impaired glucose metabolism and insulin resistance are the pathogenic factors associated with type 2 diabetes in obesity [55]. HFD-fed mice exhibit an increased body weight, linked to elevated glucose levels and reduced glucose uptake, leading to hyperlipidemia [56]. OGTT and ITT are important clinically relevant tests used to evaluate normal or impaired glucose metabolism and insulin activity in the whole body and are performed by measuring glucose levels. A previous study established a direct link between fasting and postprandial glucose levels and heightened liver enzyme levels [57]. Moreover, elevated ALT levels have been linked to triglyceride and total cholesterol synthesis [58]. The consumption of an HFD leads to abnormal plasma lipid profiles, including LDL, HDL, triglycerides, and total cholesterol, and this is associated with pivotal risk factors for coronary heart disease and atherosclerosis [59]. However, CC treatment improved glucose tolerance and insulin activity, based on the OGTT and ITT, while restoring irregular lipid profiles. These results suggest a beneficial effect of CC on impaired lipid and glucose metabolism.

The onset of obesity involves substantial weight gain and a heightened risk of developing metabolic syndrome in rodent models fed an HFD due to disturbances such as impaired insulin sensitivity, glucose tolerance, dyslipidemia, hypertension, and abdominal obesity [60]. To comprehensively assess whole-body metabolic changes, metabolic cage monitoring is used to track certain parameters, including VO_2, CO_2, the RER, and EE, while also analyzing total body fat and lean body mass. VO_2 consumption is related to lean body mass, and the levels of O_2 consumption and CO_2 expiration are related to indirect heat production, which is used to calculate EE [61]. Notably, CC significantly decreased the total body fat percentage and increased the lean body mass in HFD-fed mice through improvements in metabolic parameters. Previous studies have demonstrated that various phytochemicals can induce browning of white fat [15], increase mitochondrial biogenesis, and enhance fat oxidation [16], providing insights into their potential effects against obesity and metabolic dysfunction related to energy expenditure [15]. In this study, we observed enhanced mRNA expression levels of adipocytes browning (*Ucp1*, *Prdm16*, and *Pgc1a*), mitochondrial biogenesis (*Cytc*, and *Cox4b*) and β-oxidation associated (*Mccad*) markers in CC administered HFD induced mice, indicating increased RER, EE and lean body mass. Furthermore, these results provide additional support for potential effects of CC on inhibiting body weight gain and improving metabolic parameters during mice metabolic cage analysis. These findings suggest that CC supplementation promotes adipose tissue remodeling towards a more metabolically active state, characterized by enhanced mitochondrial function and fat oxidation, resulting in mitigating the dysregulated energy metabolism induced by the HFD. Taken together, CC effectively inhibited body weight gains by improving obesity-related parameters regardless of the varying degrees of obesity or dietary modifications during the weight-loss phase. These findings suggest that CC could be a reliable natural supplement that could be used throughout the weight-management process. However, this study underlines the requirement for additional research and clinical investigations to validate and establish the efficacy of CC as a viable therapeutic herbal medicine for obesity management. Clinical trials, in particular, are essential to assess the safety and efficacy of CC extract in human subjects, certifying potential benefits of CC in overweight management are supported by rigorous scientific evidence.

5. Conclusions

This research demonstrated the inhibitory effects of CC on adipocyte differentiation, the accumulation of lipid droplets *in vitro*, and body weight gains *in vivo*. The effects of CC were attributed to the suppression of adipogenesis, increases in lipolytic factors, and modulation of obesity-related parameters, irrespective of dietary variations or the treatment duration. Overall, this study suggests that CC might be a promising natural herbal agent for the treatment and/or prevention of obesity.

Ethics approval and consent to participate

The animal experiments were approved by the Institutional Animal Care and Use Committee of Ajou University School of Medicine (approval no. 2022–0005)

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CRediT authorship contribution statement

Je Kyung Seong: Investigation, Data curation. **Il Yong Kim:** Investigation, Data curation. **Hyesun Go:** Investigation, Data curation. **Eunkuk Park:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Seon-Yong Jeong:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Laxmi Prasad Uprety:** Writing – original draft, Supervision, Formal analysis. **Chang-Gun Lee:** Writing – original draft, Supervision, Formal analysis. **Kang-Il Oh:** Validation, Software, Methodology. **Hyesoo Jeon:** Validation, Software, Methodology. **Subin Yeo:** Validation, Software, Methodology. **Yoonjoong Yong:** Validation, Software, Methodology.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests. Seon-Yong Jeong reports financial support was provided by Ministry of Health & Welfare, Republic of Korea. Eunkuk Park reports financial support was provided by Ministry of SMEs and Startups. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Availability of data and material

The data generated and analyzed in the present study are available from the corresponding author upon a reasonable request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.biopha.2024.116799.

References

[1] P.G. Kopelman, Obesity as a medical problem, Nature 404 (6778) (2000) 635–643, https://doi.org/10.1038/35007508.
[2] V.V. Thaker, Genetic and epigenetic causes of obesity, Adolesc. Med. State Art. Rev. 28 (2) (2017) 379–405.
[3] V. Kotsis, S. Stabouli, S. Papakatsika, Z. Rizos, G. Parati, Mechanisms of obesity-induced hypertension, Hypertens. Res. 33 (5) (2010) 386–393, https://doi.org/10.1038/hr.2010.9.
[4] A. Shang, R.Y. Gan, X.Y. Xu, Q.Q. Mao, P.Z. Zhang, H.B. Li, Effects and mechanisms of edible and medicinal plants on obesity: an updated review, Crit. Rev. Food Sci. Nutr. 61 (12) (2021) 2061–2077, https://doi.org/10.1080/10408398.2020.1769548.
[5] C. Arroyo-Johnson, K.D. Mincey, Obesity epidemiology worldwide, Gastroenterol. Clin. North Am. 45 (4) (2016) 571–579, https://doi.org/10.1016/j.gtc.2016.07.012.
[6] M. Cifuentes, C. Albala, C.V. Rojas, Differences in lipogenesis and lipolysis in obese and non-obese adult human adipocytes, Biol. Res. 41 (2) (2008) 197–204, https://doi.org/10.4067/S0716-97602008000200009.
[7] J. Jakab, Adipogenesis as a potential anti-obesity target: a review of pharmacological treatment and natural products, Diabetes Metab. Syndr. Obes. 8 (2021) 67–83, https://doi.org/10.2147/dmso.s281186.
[8] M. Edwards, S.S. Mohiuddin, Biochemistry, Lipolysis, StatPearls, StatPearls Publishing Copyright © 2023, StatPearls Publishing LLC., Treasure Island (FL), 2023.
[9] L.S. Szczepaniak, E.E. Babcock, F. Schick, R.L. Dobbins, A. Garg, D.K. Burns, J. D. McGarry, D.T. Stein, Measurement of intracellular triglyceride stores by H spectroscopy: validation in vivo, Am. J. Physiol. 276 (5) (1999) E977–E989, https://doi.org/10.1152/ajpendo.1999.276.5.e977.
[10] D. Moseti, A. Regassa, W.K. Kim, Molecular regulation of adipogenesis and potential anti-adipogenic bioactive molecules, Int. J. Mol. Sci. 17 (1) (2016) 124, https://doi.org/10.3390/ijms17010124.
[11] F.B. Kraemer, W.J. Shen, Hormone-sensitive lipase: control of intracellular tri-(di-) acylglycerol and cholesteryl ester hydrolysis, J. Lipid Res. 43 (10) (2002) 1585–1594, https://doi.org/10.1194/jlr.r200009-jlr200.
[12] A. Kong, D. Xu, T. Hao, Q. Liu, R. Zhan, K. Mai, Q. Ai, Role of acyl-coenzyme A oxidase 1 (ACOX1) on palmitate-induced inflammation and ROS production of macrophages in large yellow croaker (Larimichthys crocea), Dev. Comp. Immunol. 136 (2022) 104501, https://doi.org/10.1016/j.dci.2022.104501.
[13] J. Lee, Adipose fatty acid oxidation is required for thermogenesis and potentiates oxidative stress-induced inflammation, Cell Rep. 10 (2015) 266–279, https://doi.org/10.1016/j.celrep.2014.12.023.
[14] S. de Ferranti, D. Mozaffarian, The perfect storm: obesity, adipocyte dysfunction, and metabolic consequences, Clin. Chem. 54 (6) (2008) 945–955, https://doi.org/10.1373/clinchem.2007.100156.
[15] M. Colitti, S. Grasso, Nutraceuticals and regulation of adipocyte life: premises or promises, Biofactors 40 (4) (2014) 398–418, https://doi.org/10.1002/biof.1164.
[16] T. Wood Dos Santos, Q. Cristina Pereira, L. Teixeira, A. Gambero, A.V. J, M. Lima Ribeiro, Effects of polyphenols on thermogenesis and mitochondrial biogenesis, Int. J. Mol. Sci. 19 (9) (2018) 2757, https://doi.org/10.3390%2Fijms19092757.
[17] F.R. Jornayvaz, G.I. Shulman, Regulation of mitochondrial biogenesis, Essays Biochem 47 (2010) 69–84, https://doi.org/10.1042/bse0470069.
[18] P. Seale, S. Kajimura, W. Yang, S. Chin, L.M. Rohas, M. Uldry, G. Tavernier, D. Langin, B.M. Spiegelman, Transcriptional control of brown fat determination by PRDM16, Cell Metab. 6 (1) (2007) 38–54, https://doi.org/10.1016/j.cmet.2007.06.001.
[19] Y. Li, J.-S. Park, J.-H. Deng, Y. Bai, Cytochrome c oxidase subunit IV is essential for assembly and respiratory function of the enzyme complex, J. Bioenerg. Biomembr. 38 (5) (2006) 283–291, https://doi.org/10.1007/s10863-006-9052-z.
[20] J.L. Merritt, 2nd, I.J. Chang, Medium-Chain Acyl-Coenzyme A Dehydrogenase Deficiency, in: M.P. Adam, J. Feldman, G.M. Mirzaa, R.A. Pagon, S.E. Wallace, L.J. H. Bean, K.W. Gripp, A. Amemiya (Eds.), GeneReviews((R)), Seattle (WA), 1993.
[21] E.C. Weiss, D.A. Galuska, L. Kettel Khan, C. Gillespie, M.K. Serdula, Weight regain in U.S. adults who experienced substantial weight loss, 1999-2002, Am. J. Prev. Med. 33 (1) (2007) 34–40, https://doi.org/10.1016/j.amepre.2007.02.040.

[22] J. Woodcock, FDA In Brief: FDA Requests Voluntary Withdrawal of Weight-Loss Medication After Clinical Trial Shows an Increased Occurrence of Cancer, in: M. Richards (Ed.) FDA, USA, 2020.
[23] J.B. Schroll, E.I. Penninga, P.C. Gotzsche, Assessment of adverse events in protocols, clinical study reports, and published papers of trials of orlistat: a document analysis, PLoS Med 13 (8) (2016) e1002101, https://doi.org/10.1371/journal.pmed.1002101.
[24] N. Vasudeva, Natural Products: A safest Approach for Obesity, Chin. J. Integr. Med. (2012) 473–480, https://doi.org/10.1007/s11655-012-1120-0.
[25] R. Sayeed, M. Thakur, A. Gani, Celosia cristata Linn. flowers as a new source of nutraceuticals- A study on nutritional composition, chemical characterization and in-vitro antioxidant capacity, Heliyon 6 (12) (2020) e05792, https://doi.org/10.1016/j.heliyon.2020.e05792.
[26] S. Foster, Herbal Emissaries: Bringing Chinese Herbs to the West: a Guide to Gardening, Herbal Wisdom and Wellbeing, 1992.
[27] E.F. ANDERSON, Ethnobotany of Hill Tribes of Northern Thailand. I. Medicinal Plants of Akha1, Econ. Bot. 40 (1986) 38–53, https://doi.org/10.1007/BF02858945.
[28] Y.S. Kim, J.W. Hwang, S.H. Sung, Y.J. Jeon, J.H. Jeong, B.T. Jeon, S.H. Moon, P. J. Park, Antioxidant activity and protective effect of extract of Celosia cristata L. flower on tert-butyl hydroperoxide-induced oxidative hepatotoxicity, Food Chem. 168 (2015) 572–579, https://doi.org/10.1016/j.foodchem.2014.07.106.
[29] F.I. Sultan, Chromatographic separation and identification of many fatty acids and phenolic compounds from flowers of celosia cristata l. and its inhibitory effect on some pathogenic bacteria, Aust. J. Basic. Appl. Sci. 12 (2018) 25–31, https://doi.org/10.22587/ajbas.2018.12.7.4.
[30] V. Jayanthy, Use of flowers as antimelanocyte agent against UV radiation effects, Am. J. Bio-Pharm. Biochem. Life Sci. 1 (2012).
[31] R. Fitoussi, Impact of celosia cristata extract on adipogenesis of native human CD34+/CD31− Cells, J. Cosmet. Dermatol. Sci. Appl. 3 (2013) 55–63, https://doi.org/10.4236/jcdsa.2013.33A2013.
[32] B. Saad, A Review of the Anti-obesity effects of wild edible plants in the mediterranean diet and their active compounds: from traditional uses to action mechanisms and therapeutic targets, Int. J. Mol. Sci. 24 (16) (2023) 12641, https://doi.org/10.3390%2Fijms241612641.
[33] V. Varadharaj, J. Muniyappan, Phytochemical and phytotherapeutic properties of celosia species- a review, Int. J. Pharmacogn. Phytochem Res. 9 (6) (2017) 820–825, https://doi.org/10.25258/phyto.v9i6.8185.
[34] M. Sing, S. Kumar P, B. Shrivastava, P. Reddy B, A comprehensive review of phytochemical and pharmacological overview on celosia cristata for future prospective research, Asian J. Pharm. Clin. Res. 13 (12) (2020) 21–24, https://doi.org/10.22159/ajpcr.2020.v13i12.38675.
[35] T. Wang, Q. Wu, T. Zhao, Preventive Effects of Kaempferol on High-Fat Diet-Induced Obesity Complications in C57BL/6 Mice, Biomed. Res. Int. 2020 (2020) 4532482, https://doi.org/10.1155/2020/4532482.
[36] D. Torres-Villarreal, A. Camacho, H. Castro, R. Ortiz-Lopez, A.L. de la Garza, Anti-obesity effects of kaempferol by inhibiting adipogenesis and increasing lipolysis in 3T3-L1 cells, J. Physiol. Biochem. 75 (1) (2019) 83–88, https://doi.org/10.1007/s13105-018-0659-4.
[37] M.J. Dubber, V. Sewram, N. Mshicileli, G.S. Shephard, I. Kanfer, The simultaneous determination of selected flavonol glycosides and aglycones in Ginkgo biloba oral dosage forms by high-performance liquid chromatography-electrospray ionisation-mass spectrometry, J. Pharm. Biomed. Anal. 37 (4) (2005) 723–731, https://doi.org/10.1016/j.jpba.2004.11.052.
[38] D. Gray, K. LeVanseler, P. Meide, E.H. Waysek, Evaluation of a method to determine flavonol aglycones in Ginkgo biloba dietary supplement crude materials and finished products by high-performance liquid chromatography: collaborative study, J. Aoac. Int. 90 (1) (2007) 43–53, https://doi.org/10.1093/jaoac/90.1.43.
[39] B. Marcheva, K.M. Ramsey, C.B. Peek, A. Affinati, E. Maury, J. Bass, Circadian clocks and metabolism, Handb. Exp. Pharmacol. 217 (2013) 127–155, https://doi.org/10.1007/978-3-642-25950-0_6.
[40] A.G. Oliveira, B.M. Carvalho, G. Zweig Rocha, Editorial: Pharmacological and non-pharmacological therapy for obesity and diabetes - volume II, Front. Endocrinol. (Lausanne) 14 (2023) 1252536, https://doi.org/10.3389/fendo.2023.1252536.
[41] V. Valli, K. Heilmann, F. Danesi, A. Bordoni, C. Gerhäuser, Modulation of adipocyte differentiation and proadipogenic gene expression by sulforaphane, genistein, and docosahexaenoic acid as a first step to counteract obesity, Oxid. Med. Cell. Longev. 2018 (2018) 1617202, https://doi.org/10.1155/2018/1617202.
[42] S. Kersten, Mechanisms of nutritional and hormonal regulation of lipogenesis, EMBO Rep. 2 (4) (2001) 282–286, https://doi.org/10.1093/embo-reports/kve071.
[43] U. Kintscher, R.E. Law, PPARgamma-mediated insulin sensitization: the importance of fat versus muscle, Am. J. Physiol. Endocrinol. Metab. 288 (2) (2005) E287–E291, https://doi.org/10.1152/ajpendo.00440.2004.
[44] G. Elberg, Modulation of the murine peroxisome proliferator-activated receptor γ2 promoter activity by CCAAT/enhancer-binding proteins, P27815–22, J. Biol. Chem. 275 (36) (2000), https://doi.org/10.1074/jbc.m003593200.
[45] J. Berndt, P. Kovacs, K. Ruschke, N. Kloting, M. Fasshauer, M.R. Schon, A. Korner, M. Stumvoll, M. Bluher, Fatty acid synthase gene expression in human adipose tissue: association with obesity and type 2 diabetes, Diabetologia 50 (7) (2007) 1472–1480, https://doi.org/10.1007/s00125-007-0689-x.
[46] M. KUCHARSKI, Fatty acid binding protein 4 (FABP4) and the body lipid balance, Folia Biol. 65 (2017) 181–186, https://doi.org/10.3409/fb65_4.181.
[47] I. Moreno-Indias, Impaired adipose tissue expandability and lipogenic capacities as ones of the main causes of metabolic disorders, J. Diabetes Res. 2015 (2015) 970375, https://doi.org/10.1155/2015/970375.
[48] R.B. Semwal, D.K. Semwal, I. Vermaak, A. Viljoen, A comprehensive scientific overview of Garcinia cambogia, Fitoterapia 102 (2015) 134–148, https://doi.org/10.1016/j.fitote.2015.02.012.
[49] M. Vaughan, The production and release of glycerol by adipose tissue incubated in vitro, J. Biol. Chem. 237 (1962) 3354–3358, https://doi.org/10.1016/S0021-9258(19)70821-7.
[50] K. Wunderling, J. Zurkovic, F. Zink, L. Kuerschner, C. Thiele, Triglyceride cycling enables modification of stored fatty acids, Nat. Metab. 5 (4) (2023) 699–709, https://doi.org/10.1038/s42255-023-00769-z.
[51] N. Hariri, L. Thibault, High-fat diet-induced obesity in animal models, Nutr. Res. Rev. 23 (2) (2010) 270–299, https://doi.org/10.1017/s0954422410000168.
[52] S.M. Oussaada, K.A. van Galen, M.I. Cooiman, L. Kleinendorst, E.J. Hazebroek, M. M. van Haelst, K.W. Ter Horst, M.J. Serlie, The pathogenesis of obesity, Metabolism 92 (2019) 26–36, https://doi.org/10.1016/j.metabol.2018.12.012.
[53] K.M. Jo, N.-Y. Heo, S.H. Park, Y.S. Moon, T.O. Kim, J. Park, J.H. Choi, Y.E. Park, J. Lee, Serum aminotransferase level in rhabdomyolysis according to concurrent liver disease, Korean J. Gastroenterol. 74 (4) (2019) 205–211, https://doi.org/10.4166/kjg.2019.74.4.205.
[54] V. Lala, M. Zubair, D.A. Minter, Liver Function Tests, StatPearls, Treasure Island (FL), 2023.
[55] C. Nagy, E. Einwallner, Study of In Vivo Glucose Metabolism in High-fat Diet-fed Mice Using Oral Glucose Tolerance Test (OGTT) and Insulin Tolerance Test (ITT,), J. Vis. Exp. 7 (131) (2018) 56672, https://doi.org/10.3791/56672.
[56] M.P. Czech, Insulin action and resistance in obesity and type 2 diabetes, Nat. Med. 23 (7) (2017) 804–814, https://doi.org/10.1038/nm.4350.
[57] L. Gerber, Non-alcoholic fatty liver disease (NAFLD) is associated with low level of physical activity: a population-based study, Aliment Pharmacol. Ther. 36 (8) (2012) 772–781, https://doi.org/10.1111/apt.12038.
[58] S. Saligram, E.J. Williams, M.G. Masding, Raised liver enzymes in newly diagnosed Type 2 diabetes are associated with weight and lipids, but not glycaemic control, Indian J. Endocrinol. Metab. 16 (6) (2012) 1012–1014, https://doi.org/10.4103/2230-8210.103027.
[59] J. Yang, P. Zeng, L. Liu, M. Yu, J. Su, Y. Yan, J. Ma, W. Hu, X. Yang, J. Han, Y. Duan, Y. Chen, Food with calorie restriction reduces the development of atherosclerosis in apoE-deficient mice, Biochem. Biophys. Res. Commun. 524 (2) (2020) 439–445, https://doi.org/10.1016/j.bbrc.2020.01.109.
[60] S. Moreno-Fernandez, M. Garces-Rimon, G. Vera, J. Astier, J.F. Landrier, M. Miguel, High fat/high glucose diet induces metabolic syndrome in an experimental rat model, Nutrients 10 (10) (2018) 1502, https://doi.org/10.3390/nu10101502.
[61] G.I. Lancaster, D.C. Henstridge, Body composition and metabolic caging analysis in high fat fed mice, J. Vis. Exp. 24 (135) (2018) 57280, https://doi.org/10.3791/57280.